SpecialistRoute Handover — 5. Experiments

Oct 25, 2026 · @Anas Alsirafy

All v2 results come from `PlantDoc_Main.ipynb` plus three helper scripts and one label-quality notebook. Component training, pipeline evaluation, ensembling and the test-composition analysis are complete; the label-quality review CSVs and the teacher diagnostic are not.

Experiment map

Experiment	Where	Output	Status
Data preparation (leakage removal, split, test restore)	`prepare_final_dataset.py`, `verify_clahe_dataset.py`	`Datasets/dataset`, `Datasets/reports`	Done
Component training: 8 components × 5 seeds	Main 9a–9b (`run_all`)	`Experiments/v2_<Component>/`	Done; Potato retrained on ConvNeXt-Base
Assembled pipeline, per seed	Main 8 (run by 9b)	`Experiments/v2_pipeline/`	Done
Results workbook	`collect_v2_results.py`	`Experiments/v2_all_results.xlsx`	**Re-run needed:** the last workbook predates the Potato fix
Seed ensemble + flip TTA	Main 11 (11a, 11a-2, 11b, 11c)	`v2_pipeline/probs/`, `ensemble5_plain_hard_test_*`	Done
216 vs 236 test composition	Main 14	`v2_pipeline/test_216_vs_236_*.csv`	Done

Experiment	Where	Output	Status
Label quality, frozen probe, 5-fold out-of-fold	`PlantDoc_LabelQuality.ipynb` (same code as Main 13)	`Experiments/v2_label_quality/`	Scores and validation check done; review CSVs not filled in
Teacher diagnostic on training images	Main 12	`Experiments/v2_teacher/`	Built, not run
Backbone fine-tuning check	Snippet in Known Issues	—	**Not run; do first**

Code files and notebook sections

File	Purpose
`PlantDoc_Main.ipynb`	Everything for v2: config, data, models, training, evaluation, analyses
`prepare_final_dataset.py` / `.ipynb`	Builds the final dataset and reports from the original Kaggle data
`verify_clahe_dataset.py`	Checks the stored images against a fresh CLAHE run of the originals
`collect_v2_results.py`	Gathers every result into `v2_all_results.xlsx`
`PlantDoc_LabelQuality.ipynb`	Standalone label-quality analysis (reads train/val only)
`gmp_layer.py`	Generalized-mean pooling layer from a v1 experiment; import it before loading any GMP model
`PlantDoc_Archive_Tools.ipynb`	Retired v1 cells (old audits, Copy-Paste V2, old evaluations). **Never run it in the same kernel as Main**

Sections of `PlantDoc_Main.ipynb`:

Section	Contents	Runs training?
1 Setup	Imports, GPU memory growth	No
2 Configuration	`MODEL_PATHS`, `CONFIG_BASE`, `COMPONENTS`, `apply_component`	No
3 Data	Loaders, derived folders, `check_protocol()`	No
4 Preprocessing & augmentation	Preprocess functions, CutMix/MixUp, GridMask, RandAugment	No
5 Models	Model builders, `FineTuneCallback`	No
6 Training	Callbacks, `train_single_split`	No (definitions)
7 Multi-seed runner	`run_multi_seed_experiment`, progress files	No (definitions)
8 Pipeline evaluation	`evaluate_pipeline`, theory, summaries	No (definitions)
9a / **9b**	Orchestrator / **run everything**	**9b: yes**, every component × seed
10	Manual run of one component or seed	**Yes**
11	Seed ensemble + TTA (11a defs, 11a-2 caching, 11b validation, 11c test)	Inference only
12	Teacher diagnostic on training images	Inference only
13	Label quality (copy of the standalone notebook)	Inference + CPU
14	216 vs 236 test composition	No (uses cache)

The usual way in is sections 1–8, which only define things, then the one analysis section you need.

Reproducing v2 from scratch

1. **Build the dataset:** set the paths in `prepare_final_dataset.py` (original Kaggle train/test, current CLAHE train/test, output folder; `img_size` = 224) and run it. The first run stops for the near-duplicate review: fill `is_duplicate` (`y` / `n`) in `near_duplicate_review.csv`, then run again. It must end with 236 test images.

2. **Verify:** run `verify_clahe_dataset.py` with the **raw** original folders (expected: 0 flagged).
3. **Check the protocol:** in Main, run sections 1–3 and `check_protocol()` until it prints `RESULT: OK`.
4. **Train everything:** run 9a, then 9b. It goes seed by seed (Classifier first), clears GPU memory between runs, evaluates the pipeline after each seed, and resumes where it stopped. Total: 40 training runs.
5. **Collect results:** run `collect_v2_results.py`.
6. **Ensemble:** run 11a, 11a-2 and 11b (re-run 11b until it reports 160/160 caching jobs complete), then 11c once with the variant chosen on validation.
7. **Test composition:** run section 14.
8. **Label quality (optional):** run `PlantDoc_LabelQuality.ipynb` top to bottom.

Output conventions

File	Where	Contents
`<model>_seed<s>.keras`	`v2_<Component>/`	Best validation-loss checkpoint of that seed
`<model>_seed<s>_history.csv`	`v2_<Component>/`	Per-epoch loss, accuracy and learning rate
`progress.json`	`v2_<Component>/`	One record per seed: selected epoch, `val_*` and `test_*` metrics. Seeds listed here are skipped on re-run
`config.txt`	`v2_<Component>/`	The full config of the last run; used for stale detection
`pipeline_log_seed<s>.csv`	`v2_pipeline/`	Per test image: true class, routed species, prediction, correct or not

File	Where	Contents
`pipeline_summary_seed<s>.json`, `theoretical_seed<s>.json`	`v2_pipeline/`	Measured and theoretical accuracy per seed
`probs/<split>_<Component>_seed<s>[_flip].npz`	`v2_pipeline/`	Cached softmax outputs; reused by sections 11 and 14
`ensemble_variants_val.csv`, `ensemble_size_curve_*.csv`	`v2_pipeline/`	Validation choice of the ensemble variant; accuracy vs number of seeds
`ensemble5_plain_hard_test_summary.json` / `_log.csv`	`v2_pipeline/`	The final ensemble test result
`v2_run_all_state.json`	`Experiments/`	Orchestrator log: done, failed and superseded runs
`<Component>_label_quality.csv`, `_review.csv`, `_review_sheet_p*.jpg`	`v2_label_quality/`	Scores, flagged images, contact sheets

Earlier (v1) experiments

v1 selected checkpoints on the test set and used a cleaned test set, so **none of its numbers should be reported as results**. Its design conclusions still hold, and v2 inherits them.

v1 experiment	Outcome	Carried into v2
6-class router + confidence threshold vs 7-class router with Rest	7-class wins by +18.5 pp (182 vs 142 / 236, seed 42)	Yes: 7-class router
Confidence re-routing between specialists	Negative: specialists are overconfident on foreign images (Grape ≥ 0.9 on 61.7% of non-grape images)	No
PIL vs TensorFlow image loading in evaluation	PIL bicubic cost 13–18 test images	TensorFlow loader everywhere

v1 experiment	Outcome	Carried into v2
Generalized-mean pooling (ConvGeM-NeXt idea)	Explored; not adopted	No; `gmp_layer.py` kept
Copy-Paste V2 augmentation	Ablation run in v1	Not regenerated for the v2 split
PlantVillage vs PlantDoc, best config per dataset	Large lab-to-field gap, Tomato largest	Table uses v1 PlantDoc numbers; update to v2
Theory vs measured pipeline accuracy	Matches within 0.1–0.4 pp	Yes: section 8 `theoretical_pipeline`

The v1 documents (content plan, pipeline validation process, Copy-Paste notes, literature comparison, PlantVillage comparison) still quote v1 numbers such as 84.27%. Treat them as history until they are updated.